Sciatic Nerve Conductivity is Impaired by Hamstring Strain Injuries

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ABSTRACT

The aim of this study was to assess sciatic nerve conductivity in athletes with a history of hamstring strain injuries. Twenty-seven athletes with a history of hamstring strain injuries were included in the injured group. The control group consisted of 16 uninjured participants. We measured the proximal and distal latencies and calculated the sciatic nerve conduction velocity to evaluate neuronal conductivity. The results were expressed as median values and interquartile ranges. Both proximal latency and distal latency of the injured limb in the injured group were significantly longer than those of the uninjured limb ($p < 0.05$). The nerve conduction velocity of the injured limb in the injured group was significantly lower than that of the uninjured limb ($p < 0.05$). There were no significant side-to-side differences in the control group. Sciatic nerve conductivity impairments may exist in athletes with a history of hamstring strain injuries.

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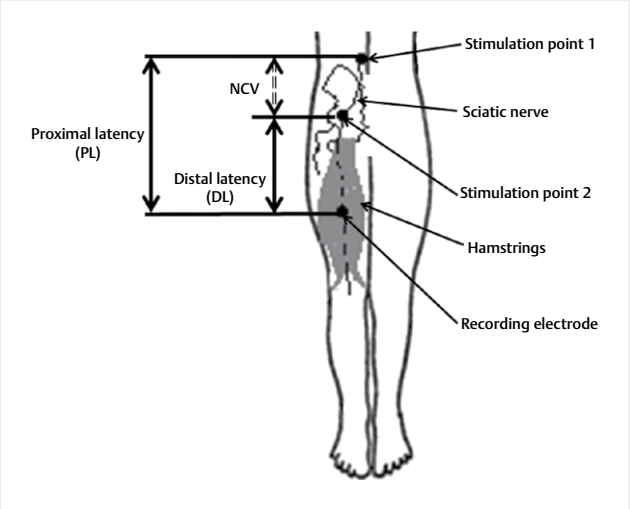
Introduction

Hamstring strain injuries (HSIs) are frequently observed in athletes who participate in strenuous sports such as rugby, American football, soccer, and track and field events [16, 20]. Decreased strength and loss of flexibility are the typical symptoms [16, 21]. In addition, HSI are characterised by high re-injury rates [2]. It is generally known that the risk factors for HSI are mainly categorized according to non-modifiable factors (age, previous HSI, and/or ethnic origin) and modifiable factors (fatigue, strength imbalances, and/or early return to sports) [2, 17, 19].

Colak et al. reported that the peripheral nerves of the upper extremities of ice hockey players are exposed to acute and chronic mechanical motions, such as the racket swing [7]. In their study, conductive time of the brachial plexus was 6 % prolonged in athletes compared with non-athlete controls. In addition, we found that nerve conductivities were impaired in animal and human models with eccentric contraction (EC)-induced muscle damage [13–15]. Degenerated nerves, thinner myelin, and decreased axon diameter were also observed after EC [13]. Further, Kami et al. reported that crush injury in the gastrocnemius muscle induced sciatic nerve injury [10]. These reports indicate that muscle injuries induced by overuse and/or strenuous exercises are possibly accompanied by functional and structural nerve responses.

A previous commentary suggested that entrapment of the lumber spinal nerves may predispose towards a high recurrence rate of hamstring injuries [18]. The lumbar nerves supply the several peripheral nerves that contract the lower limb muscles. Specifically, the hamstring and calf muscles are innervated by the sciatic nerve, which branches at the L5 and S1 regions [18]. Although these reports strongly suggest the existence of nerve impairments in muscle injuries, direct evidence of this conclusion is lacking.

The aim of this study was to investigate the motor nerve latency (M-wave) and nerve conduction velocity (NCV) of the sciatic nerve in athletes with a history of hamstring muscle strain injuries. Here, we hypothesised that sciatic nerve conductivities are impaired by a history of muscle strain injuries.



▶ **Fig. 1** Measurement of the sciatic nerve. NCV is calculated using the distance and time differences of two latencies as follows: NCV = distance/(proximal latency – distal latency). Proximal latency refers to the time of nerve root stimulation to the recording of the hamstring contraction. Distal latency is achieved by a nerve stimulation passed through the bottom of the piriformis to record the hamstring contraction. NCV, nerve conduction velocity

Materials and Methods

Before participating, the subjects received a detailed description of the study assessment method and provided informed consent. This study was performed in accordance with the International Journal of Sports Medicine [9] and the ethical committee of Nippon Sport Science University (013-H08). Forty-three collegiate students were enrolled; of them, 27 were athletes with a history of HSI defined as non-traumatic injuries of the hamstring (excluding muscle contusions) who were assigned to the injured (INJ) group; 16 uninjured participants were assigned to the control (CNT) group. In particular, subjects of the INJ group were recruited who the medical staff solicited from their respective sports clubs. The INJ group was asked about their history of HSI and return-to-sports time. The duration of return to sport was 2 weeks to 2 months. Subjects with acute and chronic strain injuries were recruited. In this study, an acute strain injury was defined as the state in which the athletes had not returned to full participation in training and were not available for selection. On the other hand, the definition of a chronic strain injury was the state in which the athletes had returned to full participation in training and games [8, 20]. Twenty-three athletes had already returned to their sports, while three subjects had not returned to play. The INJ group had a hamstring injury only once. The subjects in the CNT group had not exercised regularly in the past year but occasionally participated in recreational sports. In this study, one inclusion criterion of the CNT group specified that subjects had not suffered a past hamstring injury. All participants were asked to refrain from performing strenuous activities to eliminate the effects of acute activities. None of the subjects in either group had undergone surgery for HSI or other diseases.

The hamstring is composed of the biceps short head, which is innervated by the common fibular nerve, and the biceps long head, semitendinosus, and semimembranosus muscles, which are innervated by the tibial nerve. Both nerves are branches of the sciatic nerve, which originates at the lumbosacral plexus. Therefore, we targeted the sciatic nerve for the nerve conduction assessment.

The nerve function was examined to determine nerve conduction impairments due to neuropathy and traumatic injury [3]. The NCV was indicated as the conductive speed of the action potential per second in this study. It was calculated using the distance and time differences of two M-wave latencies, proximal latency (PL) and distal latency (DL). The PL is the conductive time from the initiation site of the sciatic nerve to the hamstring, and the DL is detected from the inferior division of the piriformis muscle to the hamstrings. The action potential was induced by stimulating two points (lower lumbar region and gluteal region) as described previously [3]. The proximal stimulation focused on the L5 root at the midpoint between the L5 spinal process and the posterior iliac crest; the distal stimulation was placed over the sciatic nerve at the gluteal fold, which is located at the midpoint between the ischial tuberosity and the greater trochanter of the femur [3] (▶ **Fig. 1**). Both latencies were detected as the compound muscle action potential detected from the hamstring contractions.

The stimulation device used a magnetic stimulator (Magstim Company Ltd., Whitland, UK). A 70-mm-diameter figure eight-shaped coil was used for the stimulation. Intensities were set from 1.6 to 2.0 Tesla to the supramaximal stimulus. The measurement positions of all subjects were standardised during the NCV testing (▶ **Fig. 1**). A recording electrode was placed at 50 % of the hamstring length, consisting of the biceps femoris, semimembranosus, and semitendinosus muscles. A reference electrode was affixed to the tendon of the semitendinosus muscle. The room temperature was maintained at 26–28 °C to exclude temperature changes in the M-wave latencies and NCV value. The NCV value reliability was estimated as the coefficient of variation of 2.1 %–3.5 % obtained from four healthy male volunteers.

In this study, the INJ group consisted of chronic and acute cases. Therefore, magnetic resonance imaging (MRI; 0.3 Tesla; AIRIS II; Hitachi, Ltd., Tokyo, Japan) was used to observe the structural change of the injured site. The signal region was confirmed based on the short inversion time inversion recovery in the coronal and axial planes [8]. The obtained images were divided into signal findings and no signal in accordance with the observation of an experienced orthopaedist.

According to the Shapiro-Wilk test, the data did not have a normal distribution, except for the physical characteristics. Therefore, nerve functions were expressed as median values and interquartile ranges. The statistical analysis used non-parametric tests. The Wilcoxon's signed-rank test was used to compare the intra-subject differences (CNT, right vs. left; INJ, uninjured vs. injured) in the NCV, PL, and DL. To compare the M-wave latencies and NCV of the CNT and INJ groups, relative side-to-side differences were calculated. In the CNT group, NCV was obtained by dividing the lower value by the higher value in each participant. Relative side-to-side differences in the PL and DL were calculated by dividing the higher value by the lower value in each participant. To obtain the relative values of the INJ group, the value of the injured limb was divided by that of the uninjured limb in each participant.

The Mann-Whitney U test was used to compare the inter-subject differences. The NCV and other variables were analysed using

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the non-parametric Spearman's product-moment correlation. Analyses were conducted using SPSS ver. 22.0 for Mac (IBM Japan, Tokyo, Japan), and the significance level was set at $p < 0.05$. Effect sizes (ES) and statistical power were calculated according to Cohen's d effect sizes [6]. Physical characteristics are expressed as means and standard deviations (SDs).

Results

The physical characteristics showed no significant differences between the CNT (age, 19.6 ± 1.4 years; height, 167.7 ± 7.0 cm; weight, 61.9 ± 7.3 kg) and INJ (age, 19.9 ± 1.1 years; height, 170.4 ± 9.2 cm; weight, 72.7 ± 16.0 kg) groups.

In the CNT group, no significant side-to-side differences were observed for the PL and DL (▶ **Table 1A**). In the INJ group, significant side-to-side differences were found for PL and DL. (▶ **Table 1B**). Further, relative side-to-side differences of the PL in the INJ group were higher than in the CNT group (CNT, 3 %; INJ, 13 %; ES = 0.78; 95 % CI = 0.13–1.41; $p < 0.05$; ▶ **Table 1C**). However, there was no significant difference in the DL between the CNT and INJ groups.

In the CNT group, the NCV of both legs showed almost similar median values, such as 72.7 m/s (right) and 72.4 m/s (left). Conversely, the NCV of the injured limb [64.5 m/s (51.7 m/s–74.3 m/s)] was significantly lower than that of the uninjured limb [76.4 (71.6–84.2)] (ES = 0.87, 95 % CI = 0.3–1.42; $p < 0.05$; post hoc power: 99 %, ▶ **Table 2**, **Fig. 2**). When comparing the INJ group with the CNT group, the relative side-to-side NCV differences in the INJ group were significantly higher than in the CNT group (CNT group, 98 %; INJ group, 84 %; ES = 2.04; 95 % CI = 1.26–2.75; $p < 0.05$; post hoc power: 98 %; ▶ **Table 2**).

High signal intensity on MRI was observed in 10 of the 27 subjects (37 %) in the INJ group. There was no significant correlation in the NCV values between the two groups with or without high signal intensity (rho, –0.124; $p = 0.539$).

The time after injury was 1–70 months. There was no significant association between the nerve functions (rho, –0.267; $p = 0.178$). Therefore, such respective variables were not correlated, except for the PL and DL. The physical characteristics, MRI observations, sport backgrounds, and nerve functions of the patients in the INJ group are shown in ▶ **Table 3**.

Discussion

This study explored whether sciatic nerve conductivity is impaired in athletes with a history of hamstring strain injuries. To assess nerve functions, we measured sciatic NCV and latencies in the CNT and INJ groups and results were compared between both groups. For the injured group, prolonged M-wave latencies and decreased NCV of the sciatic nerve were observed in the injured limb compared to uninjured side. There was no significant difference in the side-to-side comparison in the CNT group. In addition, there were significant inter-group differences in the relative side-to-side differences between the CNT and INJ groups, regardless of similar physical characteristics. Side-to-side NCV differences were observed in all of the athletes who experienced hamstring strain injuries in the INJ group.

▶ **Table 1A** Proximal and distal latencies of the subjects in the CNT group

		PL	DL
CNT group (n = 16)	Right limb (m/s)	4.43 (4.05–4.92)	2.45 (1.44–3.10)
	Left limb (m/s)	4.48 (4.04–4.79)	2.25 (1.44–2.86)

All data are expressed as medians (interquartile ranges). CNT, control; PL, proximal latency; DL, distal latency

▶ **Table 1B** Proximal and distal latencies of the subjects in the INJ group

		PL	DL
INJ group (n = 27)	Uninjured limb (m/s)	5.55 (4.15–7.13)	3.60 (1.78–4.28)
	Injured limb (m/s)	5.85 (4.45–7.87) *	3.60 (2.10–4.48) *

All data are expressed as medians (interquartile ranges). INJ, injured; PL, proximal latency; DL, distal latency; * $p < 0.05$ vs. the uninjured limb

▶ **Table 1C** Percent increase in the proximal and distal latencies of the subjects in the CNT and INJ groups.

	PL	DL
CNT group (%)	3 (1–6)	5 (1–13)
INJ group (%)	13 (8.5–19.2)†	7 (1–14)

All data are expressed as medians (interquartile ranges). CNT, control; INJ, injured; PL, proximal latency; DL, distal latency; † $p < 0.05$ vs. the CNT group

▶ **Table 2** Sciatic NCV among subjects in the CNT and INJ groups.

CNT group (n = 16)	Right limb (m/s)	72.7 (64.8–79.1)
	Left limb (m/s)	74.0 (64.6–78.8)
	Left/right (%)	98
INJ group (n = 27)	Uninjured limb (m/s)	76.4 (71.6–84.2)
	Injured limb (m/s)	64.5 (51.7–74.3) *
	Injured/uninjured (%)	84†

All data are expressed as medians (interquartile ranges). CNT, control; INJ, injured; NCV, nerve conduction velocity; † $p < 0.05$ vs. the CNT group, * $p < 0.05$ vs· the uninjured limb

PL, DL, and NCV measurements are used to assess nerve function in a variety of temporal nerve functional impairments and injuries [3, 7, 11]. In this study, the PL was significantly longer in the INJ group (▶ **Fig. 2**). We previously found longer latency in the musculocutaneous nerve after EC of the biceps brachii in humans [14]. We also confirmed that EC reduced maximal voluntary elbow flexor torque and muscle pain. These results indicated that muscle damage might be associated with increased PL and decreased NCV. When taken together, these findings show that sciatic nerve function was affected by hamstring strain injuries. Although there were no significant relative side-to-side differences in the DL between the INJ and CNT groups, the data distributions showed that the DL in the INJ group was longer than in the CNT group. In the latter half of the discussion section, we address the reason why only the PL, but not the DL, showed a significant difference.

Since M-wave latency includes axonal conductivity, neuromuscular transmission, and electrical conductivity in the muscle, we calculated NCV to focus on axonal conductivity. We found that NCV

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▶ **Table 3** Physical characterisrics, MRI findings, and nerve conductivity of the subjects in the INJ group.

Subject (No.)	Sex	Time after injury (months)	Hamstring flexibility (°) (Uninjured/Injured)	MRI signal	Sport	PL (m/s) (Uninjured/ Injured)	DL (m/s) (Uninjured/ Injured)	NCV (m/s) Uninjured/ Injured)
1	Male	5	75/80	yes	American football	8.82/10.40	6.42/6.42	75.0/45.2
2	Male	6	85/85	yes	American football	4.15/5.95	1.61/3.09	72.7/64.5
3	Male	2	85/85		American football	4.18/5.00	1.85/2.25	85.7/72.7
4	Male	6	90/90		American football	6.91/7.66	4.40/4.90	83.6/76.1
5	Male	24	90/90	yes	Rugby	5.91/6.79	3.60/3.90	77.6/62.3
6	Male	9	80/65		Rugby	7.35/8.07	4.60/4.95	76.4/67.3
7	Male	1	70/80		Rugby	3.90/4.20	1.69/1.70	95.0/84.0
8	Male	12	85/70	yes	Rugby	4.75/4.20	1.65/1.95	84.4/61.2
9	Male	18	90/90		Rugby	4.15/4.20	1.70/1.70	81.6/72.8
10	Male	2	90/90	yes	Rugby	8.05/8.60	5.48/5.85	76.0/69.1
11	Male	1.5	90/90	yes	Rugby	4.20/4.45	2.25/2.50	84.4/76.0
12	Male	70	80/80		Rugby	4.45/5.85	2.19/3.35	84.0/76.0
13	Male	6	90/90		Rugby	6.75/7.30	4.25/4.50	68.0/60.7
14	Male	8	80/90		Track and field	5.55/5.85	3.60/3.60	92.3/80.0
15	Male	18	90/80		Track and field	3.73/4.15	1.45/1.65	71.1/64.0
16	Male	12	90/90	yes	Track and field	3.79/5.00	1.60/1.70	50.0/34.5
17	Female	18	90/90		Track and field	3.90/4.45	1.95/2.25	91.0/81.8
18	Female	18	90/105	yes	Track and field	4.09/4.45	1.89/1.65	95.5/75.0
19	Male	16	90/90		Track and field	8.59/9.70	4.45/4.45	45.8/36.2
20	Female	14	90/80	yes	Track and field	5.25/5.85	2.80/2.76	73.5/58.1
21	Male	36	90/90		Track and field	6.40/7.60	3.90/4.00	76.0/52.8
22	Female	16	80/80		Track and field	8.25/9.70	4.30/4.75	45.5/30.3
23	Male	60	80/80		Track and field	6.55/7.45	4.05/4.70	72.0/65.5
24	Male	18	90/70		Track and field	3.60/3.85	1.40/1.41	81.1/73.5
25	Male	3	90/90		Sepak takraw	6.85/8.30	4.10/4.15	76.4/50.6
26	Female	1	120/120		Sepak takraw	7.65/9.40	3.85/3.90	50.0/34.5
27	Male	11	90/80	yes	Soccer	10.50/10.15	4.45/4.45	55.4/32.5

MRI, magnetic resonance imaging; INJ, injured; PL, proximal latency; DL, distal latency; NCV, nerve conducrtion velocity

of the injured limb was significantly decreased compared to that of the uninjured side (ES = 0.87; ▶ **Table 2**). We further confirmed that side-to-side differences of the sciatic NCV in HSI athletes were significantly greater than the side-to-side differences in the CNT group (ES = 2.04; ▶ **Table 2**). We previously observed a significant 21 % decrease in sciatic NCV that was indicated from between the origin of the sciatic nerve to the branching point to the MG after gastrocnemius EC in rats [13]. Although we cannot rule out functional impairments in the neuromuscular junction and electrical conductivities of the muscle, we believe that nerve conductive impairments are associated with HSI.

Our recent studies found the dominant nerve impairment after unaccustomed EC [13–15]. NCV deficit (a single EC bout: 21 %; multiple EC bouts: 42 %) and prolonged M-wave latency (a single EC bout: 24 %) were induced in both animal and human studies [13–15]. In this study, we found an 11 % prolonged PL and 10–30 % reduction of sciatic NCV in athletes with a history of HSI, suggesting that the reduction rate in strain injury was comparable with that in EC-induced nerve impairment. Unaccustomed EC during high-velocity sprinting and sudden accelerations are major risk factors for HSI [1]. Therefore, the present results might support previous observations showing that nerve damage was possibly induced by EC.

However, it is unclear whether nerve conductivity impairment was caused by actual nerve tissue damage. In our previous study, macrophage invasion and loss of myelin protein were also caused by a single bout of EC [15]. In addition, myelin sheath and axonal thinning were observed on electron microscopy after repetitive EC [13]. In the present study, impaired nerve conductivity was sustained for months or years, although some subjects had already returned to play. Therefore, we think that the persistent nerve conductivity impairment is due to the actual nerve tissue damage, such as the myelin sheath and/or axonal thinning.

In this study, we measured the sciatic NCV between the origin of the sciatic nerve at the lumbosacral plexus and the piriformis area in which the NCV was calculated from the PL and DL. The NCV and PL were significantly impaired in the INJ group; however, the relative side-to-side difference in the DL was not significant. Chang et al. reported that retrograde axonal atrophy was confirmed by measuring the NCV in patients with carpal tunnel syndrome (CTS) [4]. They observed significant reductions in the wrist-palm motor conduction velocity (36.2 %) as the typical symptom of CTS. In ad-

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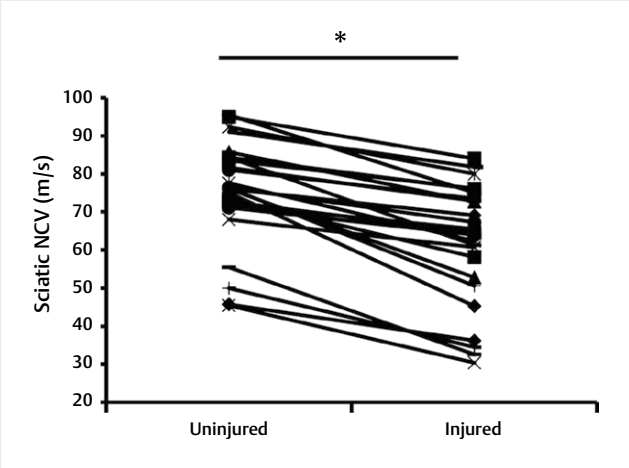
dition, the median forearm motor conduction velocity (FMCV) also significantly decreased compared with the normal control group (4.43 %). In the previous study, the FMCV, rather than the wrist-palm motor conduction velocity, was measured at a distal site. Therefore, if the initial nerve conductivity impairment occurs near the exercised muscle, the effect of the hamstring muscle injury is propagated to the remote region of the sciatic nerve. Kami et al. [10] reported that a contusion injury to the gastrocnemius muscle induces a sciatic nerve injury and that the damage occurs in a retrograde fashion toward the lumbar spinal nerve. Similarly, the dying-back phenomenon is also known as a retrograde nerve degeneration process that is observed in neuronal disorders, such as Charcot-Marie-Tooth and amyotrophic lateral sclerosis [5, 22]. Hence, we think that sciatic nerve conductivity impairment initially occurs near the injured site of the hamstring and that the impairment might propagate toward the spinal cord.

Nerve conductive function is assessed using electrical stimulation [7, 12]. However, electrical stimulation is invasive because the sciatic nerve is located deep beneath the skin. On the other hand, magnetic stimulation is a beneficial tool that enables direct percutaneous stimulation of the sciatic nerve [3]. Therefore, we used a pulsed magnetic field to induce electrical activity in the sciatic nerve. Chang et al. [3] measured sciatic NCV by magnetic field stimulation and reported that the normal sciatic NCV was 68.7 ± 10.1 m/s. In this study, the NCV value of the CNT group was 72.3 ± 12.3 (right) and 71.5 ± 12.3 (left), values that were similar to the value reported by Chang et al. They also found that the sciatic NCV was decreased by 19 % in patients with piriformis syndrome [3]. We also found a similar decrease of 14 % in the injured limb. These lines of evidence suggest that magnetic field stimulation is an applicable substitute for electrical stimulation for measuring sciatic NCV.

This study has several limitations. First, it is unclear whether the significant differences in the NCV and PL are reliable because the sample size was small in the present study. However, we analysed the effect size of the NCV and latency as follows: PL: CNT group vs. INJ group; ES = 0.78; 95 % CI = 0.13–1.41; NCV: CNT group vs INJ group ES = 2.04; 95 % CI = 1.26–2.75. Although the sample size was small, the relatively large effect sizes indicate that the results obtained in this study are reliable. Second, we did not examine the injured site and damage degree of the HSI. There may be differences in the M-wave latencies and NCV characteristics if the damaged areas are different. However, we found that all athletes who had hamstring strain injuries showed side-to-side differences in the sciatic NCV. Thus, we think that sciatic NCV would be affected by HSI regardless of injured sites and severity. Thus, we think that sciatic NCV would be affected by HSI regardless of injured sites and severity. Third, all participants in the INJ group were athletes, whereas those in the CNT group sometimes participated in recreational sports. Hence, the observed differences in nerve function may be because of the different activity levels of the participants. Because we observed that nerve function was almost similar among the uninjured limbs of all participants, we think that the participants in the CNT group serve as a suitable reference for the INJ group. Finally, a longitudinal examination is necessary to clarify the contribution of nerve impairments to HSI.



▶ **Fig. 2** NCV values between the uninjured and injured limbs in the INJ group. The NCV values significantly decrease in the injured limb (* $p < 0.05$). Slower NCV values in the injured limb are observed in all examined injured subjects. NCV, nerve conduction velocity; INJ, hamstring strain injuries group

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Conclusion

In conclusion, sciatic nerve conductivity is impaired in athletes with a history of HSI. We suggest that sciatic nerve disorders might be induced by strain injuries owing to the eccentric movements of the hamstrings. This is the first study to show that sciatic nerve conductivity impairment is observed in athletes with a history of HSI.

Conflict of interest

The author have no conflict of interest to declare.

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